

Packages to Install

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Install packages for the 2026 demos and labs

Run the chunk below once before working through the 2026 SISBID unsupervised learning demos and labs. The code checks which packages are already installed and only installs the missing packages.

```
options(repos = c(CRAN = "https://cran.rstudio.com"))

cran_packages <- c(
  "cluster",
  "fastICA",
  "GGally",
  "ggplot2",
  "ggrepel",
  "glasso",
  "glmnet",
  "huge",
  "igraph",
  "ISLR",
  "ISLR2",
  "knitr",
  "mclust",
  "NMF",
  "pheatmap",
  "Rtsne",
  "rmarkdown",
  "sda",
  "umap"
)

install_if_missing <- function(pkgs) {
  is_installed <- vapply(pkgs, requireNamespace, logical(1), quietly = TRUE)
  missing_pkgs <- pkgs[!is_installed]

  if (length(missing_pkgs) == 0) {
    message("All requested packages are already installed.")
  } else {
    message("Installing: ", paste(missing_pkgs, collapse = ", "))
    install.packages(missing_pkgs)
  }
}

install_if_missing(cran_packages)
```

Package list by file

File	Packages used
2026_SISBID_PCA_Demo.Rmd	ggplot2, GGally, ISLR, ISLR2, knitr
2026_SISBID_PCA_Lab.Rmd	ggplot2, fastICA, NMF, umap, Rtsne, GGally, knitr
2026_SISBID_Clustering_Demo.Rmd	ggplot2, cluster, ISLR, ISLR2, knitr
2026_SISBID_Clustering_Lab_fixed_v2.Rmd	ggplot2, NMF, cluster, mclust, ggrepel, pheatmap, umap, knitr
2026_SISBID_Testing_Demo_fixed.Rmd	sda, ggplot2, knitr
2026_SISBID_Graphical_Models_Demo.Rmd	igraph, huge, glasso, glmnet, ggplot2, knitr
2026_SISBID_Unsupervised_Lab_v3.Rmd	ggplot2, GGally, igraph, umap, Rtsne, pheatmap, huge, knitr

Notes

- ISLR and ISLR2 are both listed because the PCA and clustering demos try to use the older ISLR data package first and then fall back to ISLR2 if needed.
- `cluster` is commonly installed with R as a recommended package, but it is listed here because the clustering demo/lab uses silhouette diagnostics from `cluster`.
- `knitr` and `rmarkdown` are included because these files are R Markdown documents and some demos use `knitr::kable()` for tables.
- Bioconductor packages from older package lists were removed because they are not used by the final 2026 files in this collection.